FMP Module 2 — Fundamentals of Economics: Workbook Answers

Each question is quoted verbatim from the workbook (V06, 25-05-19), followed by a model answer grounded in the Module 1 and Module 2 study guides. Total marks: 110.

QUESTION 1 · Foundational Concepts [15 marks]

1.1 What is the central concept in economics that explains the fundamental economic problem, and how does it affect decision-making? *(3)*

ANSWER

Scarcity is the central concept of economics. It captures the fact that human wants are unlimited while resources — money, time, labour, raw materials, land — are finite.

» Forces every actor (household, firm, government) to **choose** between competing uses.

» Every choice carries an **opportunity cost** — the value of the next-best alternative foregone.

» Drives the three core economic questions: **what** to produce, **how** to produce it, and **for whom**.

1.2 What is the difference between explicit costs and implicit costs in a company's financial analysis? *(3)*

ANSWER

	Explicit costs	Implicit costs
Cash exchange?	Yes — actual outflows	No cash changes hands
Measurable?	Yes — recorded in accounts	Difficult to measure exactly
Examples	Wages, rent, raw materials, interest paid	Owner's unpaid labour, interest foregone on own capital, opportunity cos

Accounting profit deducts only explicit costs; **economic profit** deducts both.

1.3 How is opportunity cost calculated, and why is it important in economic decision-making? *(3)*

ANSWER

Opportunity cost = value of the next-best alternative foregone. In investment terms it is the return on the most profitable opportunity that was rejected, compared to the return on the option actually chosen.

Importance:

» Resources are scarce — capital should flow to its highest-value use.

» Forces honest comparison of alternatives rather than evaluating a choice in isolation.

» Underpins every consumer, investment, business and policy decision (e.g. the 2010 ten-thousand-bitcoin pizza, valued at ~US$41 then and over US$690m by 2024).

1.4 Explain the concept of market price and the factors that influence its determination. *(3)*

ANSWER

Market price is the price at which **quantity supplied = quantity demanded** (market equilibrium). It is set by the interaction of supply and demand, not by any single party.

Factors that influence it:

» **Demand side:** consumer income, tastes, expectations, prices of substitutes & complements, population.

» **Supply side:** input costs, technology, taxes / subsidies, number of producers, expectations.

» **External:** regulation, exchange rates, weather (commodities), political risk.

1.5 What is the difference between opportunity cost and risk in economics? *(3)*

ANSWER

	Opportunity cost	Risk
What it measures	Value of the alternative given up	Possibility that actual return differs from expected; potential los
When it bites	At the moment of choice — comparing options	After the choice is made — uncertainty within the chosen optic
Certainty	Knowable relative to the decision	Probabilistic / unknown ex-ante

QUESTION 2 · Macro vs Micro [9 marks]

2.1 What is the primary focus of macroeconomics, and how does it differ from microeconomics? *(3)*

ANSWER

Macroeconomics studies the economy as a whole — aggregates such as GDP, inflation, unemployment, interest rates, exchange rates and government budgets. It evaluates **monetary & fiscal policy** and the business cycle.

Microeconomics studies **individual** agents — households, firms and specific markets — and how they make choices over scarce resources (supply, demand, prices, elasticity, market structures).

Macro = the forest (whole economy); Micro = the trees (individual actors).

2.2 Why are macroeconomic factors significant for investors, and how do they impact investment portfolios? *(3)*

ANSWER

Macro factors set the **backdrop** against which all asset prices move:

» **GDP growth** drives corporate earnings → equity returns.

» **Interest rates & inflation** drive bond yields and the discount rate used to value all assets.

» **Currency moves** translate offshore earnings and rebase imported costs.

» **Policy stance** (fiscal expansion / contraction) alters risk premia and sector rotation.

Portfolio impact shows up in **asset allocation** (equity vs bonds vs cash), **duration**, **sector tilts**, and **geographic / currency exposure**.

2.3 What are the key principles studied in microeconomics, and how do they relate to human decision-making? *(3)*

ANSWER

» **Rational choice & utility maximisation** — agents pick the option giving the most satisfaction within their budget.

» **Marginal analysis** — decisions are made at the margin (one extra unit), not on totals.

» **Law of supply & demand** — prices coordinate buyers and sellers; surpluses and shortages self-correct.

» **Elasticity** — how strongly demand or supply responds to price changes; informs pricing & taxation.

» **Market structures** (perfect competition, monopolistic, oligopoly, monopoly) — shape pricing power and consumer welfare.

These principles model the trade-offs households and firms face every day: buy or save, work or leisure, produce locally or import, invest in project A or project B.

QUESTION 3 · Positive vs Normative [9 marks]

3.1 What is the difference between positive economics and normative economics? *(3)*

ANSWER

	Positive	Normative
Question asked	What is?	What should be?
Nature	Descriptive, fact-based, testable	Prescriptive, value-laden, opinion-based
Example	“CPI inflation in SA is 5.2%”	“Inflation is too high — SARB must hike”

3.2 What are the key characteristics of positive economics? *(3)*

ANSWER

- » **Objective** — independent of personal preferences or value judgements.
- » **Empirical** — claims are testable against data and can be falsified.
- » **Model-driven** — uses theory (e.g. supply & demand) to derive cause-and-effect relationships.
- » **Predictive** — generates if-then statements that can be checked in the real world.

3.3 How does normative economics relate to behavioural economics? *(3)*

ANSWER

Normative economics prescribes what people *ought* to do (rational, utility-maximising). **Behavioural** economics studies what people *actually* do — incorporating cognitive biases, heuristics and emotion.

- » Behavioural findings (loss aversion, anchoring, herding, framing) reveal where real behaviour **departs** from the rational ideal.
- » This challenges classical normative prescriptions and informs **policy & product design** — nudges, default opt-ins, simplified disclosures.
- » Together they shape modern welfare economics: design systems that improve outcomes given how humans actually decide.

QUESTION 4 · Demand & Supply [24 marks]

4.1 Explain consumer choice theory and its relationship to demand theory. *(4)*

ANSWER

Consumer choice theory says consumers seek to maximise their **utility** (satisfaction) subject to a budget constraint. They allocate spending so the **marginal utility per rand** is equal across all goods.

- » **Diminishing marginal utility:** each extra unit of a good adds less satisfaction than the one before.
- » So consumers are only willing to buy more if the **price falls**.
- » This produces a **downward-sloping demand curve** — the foundation of demand theory.
- » Consumer choice → individual demand → market demand.

4.2 What is the difference between a change in demand and a change in quantity demanded? *(4)*

ANSWER

	Change in quantity demanded	Change in demand
Cause	Price of the good itself	Any non-price factor
Effect on curve	Movement ALONG the curve	SHIFT of the whole curve
Examples	Petrol price rises → fewer litres bought	Income rises → demand for cars shifts right

Distinguishing the two is essential for correctly identifying new equilibria after a shock.

4.3 How does the demand curve typically behave, and what does it represent? *(4)*

ANSWER

» Slopes **downward** left-to-right — an **inverse relationship** between price and quantity demanded.

» Shows the maximum quantity consumers are willing & able to buy at each price, holding all else constant (ceteris paribus).

» Price is plotted on the **Y-axis** as the independent variable — an exception to standard maths convention.

» Underlying causes: diminishing marginal utility, the substitution effect (switch to cheaper alternatives) and the income effect.

4.4 What factors can cause a shift in the demand curve? *(4)*

ANSWER

» **Income** — rises in disposable income increase demand for normal goods, decrease it for inferior goods.

» **Tastes & preferences** — fashion, health trends, advertising.

» **Prices of related goods** — substitutes (rival products) and complements (paired products).

» **Expectations** — anticipated price changes or income shocks pull purchases forward or push them back.

» **Number of buyers** — population growth, demographic shifts, new market access.

» **Government policy** — taxes, subsidies, regulation that change effective price.

4.5 What is demand elasticity, and how does it impact businesses? *(4)*

ANSWER

Price elasticity of demand (PED) = % change in quantity demanded ÷ % change in price.

» **PED > 1 — elastic:** quantity highly responsive; price cuts *raise* revenue.

» **PED < 1 — inelastic:** quantity barely changes; price hikes *raise* revenue (e.g. fuel, basic foods).

» **PED = 1 — unit elastic:** revenue unchanged.

Business impact: guides pricing strategy, tax incidence forecasting, promotional design, and resilience modelling — necessities can absorb price pressure; luxuries cannot.

4.6 Describe the law of supply and how it is represented by the supply curve. *(4)*

ANSWER

Law of supply: all else equal, the quantity producers are willing to supply **rises** as the price rises, and falls as the price falls.

» Driven by **profit incentive** (higher price → higher margin) and **increasing marginal cost** (extra units cost more to produce).

» Represented by an **upward-sloping** supply curve.

» **Movement along** the curve = price-only change; **shift** of the curve = costs, technology, taxes, number of producers, expectations.

» Together with the demand curve, the supply curve fixes the **market equilibrium** price and quantity.

QUESTION 5 · Fiscal Policy [12 marks]

5.1 Discuss the advantages of fiscal policy. *(6)*

ANSWER

» **Direct & powerful** impact on aggregate demand via government spending and tax changes — particularly effective at the **zero lower bound** where monetary policy is exhausted.

» **Targeted** — can be channelled at specific sectors, regions or groups (infrastructure, health, social grants) rather than across the whole economy.

» **Automatic stabilisers** (UIF, progressive tax, unemployment grants) smooth shocks without new legislation.

» **Long-run capacity** — productive investment in infrastructure, education and R&D raises potential GDP, not just current demand.

» **Distributional** — can address inequality and poverty directly through tax-and-transfer choices.

» **Democratically accountable** — every line item passes through Parliament's budget process, giving transparency.

5.2 Discuss the disadvantages of fiscal policy. *(6)*

ANSWER

» **Political delays** — budget cycles are long; legislation can take months, by which time the shock has passed.

» **Debt accumulation** — persistent deficits raise the debt-to-GDP ratio and threaten fiscal sustainability.

» **Crowding-out** — heavy government borrowing pushes up bond yields, raising the cost of capital for private investment.

» **Implementation lags** — even once approved, infrastructure spend can take years to land in the real economy.

» **Risk of inefficiency & political capture** — spending may chase votes rather than economic return.

» **Sovereign / currency consequences** — credit-rating downgrades, capital flight and currency weakness if markets lose confidence in the fiscal path.

QUESTION 6 · The Business Cycle [15 marks]

6.1 Identify the phase of economic activity being experienced at each point on the business cycle graph and complete the table. *(4)*

ANSWER

Position	Phase of Economic Activity
Point A	Trough — lowest point of the cycle; output below long-term trend; high unemployment.
Points A to B	Expansion / Recovery — output rising back towards and above potential.
Point B	Peak — cycle high; economy at or above potential; capacity constraints & inflation pressure.
Points B to C	Contraction / Recession — output falling; rising unemployment; falling demand.

6.2 Complete the table below with the most likely direction of economic indicators during the identified phases. *(3)*

ANSWER

Indicator	Points A → B (Expansion)	Points B → C (Contraction
Gross Domestic Product	↑ Rising — output expanding above trend	↓ Falling — output c
Unemployment	↓ Falling — hiring picks up as demand grows	↑ Rising — retrench
Consumer Price Index	↑ Rising — demand pulls prices up, often peaking late in expansion	↓ Falling / moderating</

6.3 If South Africa's economic activity was at a point moving from B to C on the above graph, how would you expect investments in equities and bonds to be performing? *(8)*

ANSWER

Movement from **B to C is a contraction / recession** — output falling, unemployment rising, inflation moderating, central bank typically **cutting** rates to support growth.

Equity markets — poor performance overall:

» Corporate **earnings fall** as demand and pricing power weaken → equity prices decline.

» **Risk premium widens** → multiples compress; investors rotate **out of cyclicals** (banks, industrials, discretionary) **into defensives** (consumer staples, healthcare, utilities, gold).

» Dividends may be cut; smaller caps and highly geared companies hit hardest.

Bond markets — typically strong performance:

» Falling rates & rate-cut expectations push **bond yields down** → bond **prices up** (inverse relationship).

» **Flight to safety** drives flows into government bonds, supporting long-duration sovereigns most.

» **Credit spreads widen**, so investment-grade tends to outperform high-yield / junk; weak credits face higher default risk.

» Net: a contraction is generally **bad for equities, good for high-grade duration** — the textbook case for a defensive equity / long-bond rotation late in the cycle.

QUESTION 7 · CPI Calculation [8 marks]

7.1 Calculate the cost of this basket in 2023 and 2024. Show your calculations. *(4)*

ANSWER

Good	Qty	2023 Price	2023 Cost	2024 Price	2024 Cost
Chickens	48	12.50	600.00	12.90	619.20
Bread	120	1.15	138.00	1.25	150.00
Soccer Tickets	18	45.00	810.00	46.00	828.00
BASKET TOTAL			1 548.00		1 597.20

2023 basket = (48 × 12.50) + (120 × 1.15) + (18 × 45.00) = 600 + 138 + 810 = **1 548.00**

2024 basket = (48 × 12.90) + (120 × 1.25) + (18 × 46.00) = 619.20 + 150.00 + 828.00 = **1 597.20**

7.2 Using your 7.1 results, calculate a consumer price index with 2023 as the base year. Show your calculations. *(2)*

ANSWER

CPI_{year} = (Cost of basket in year ÷ Cost of basket in base year) × 100

» **CPI 2023 = (1 548.00 / 1 548.00) × 100 = 100.00** (base year)

» **CPI 2024 = (1 597.20 / 1 548.00) × 100 = 103.18**

7.3 Calculate the rate of inflation in Country ABC in 2024. Show your calculations. *(2)*

ANSWER

Inflation rate = $(CPI_{2024} - CPI_{2023}) \div CPI_{2023} \times 100$

= (103.18 – 100.00) / 100.00 × 100 = **3.18%**

QUESTION 8 · US Retail Sales Extract [5 marks]

8.1 Based on your knowledge of the business cycle, what stage of the economic cycle does this country appear to be in? *(1)*

ANSWER

Expansion — likely late expansion / approaching the peak. The 6.2% annual jump in retail sales is the strongest in four years, with durables rising 1.7% in a single month — consistent with strong consumer demand and a maturing upswing.

8.2 How would you expect the equity markets to have performed during the calendar year referenced? *(2)*

ANSWER

Equity markets would likely have **performed strongly**:

» Strong consumer spending → strong corporate earnings, especially in consumer discretionary & durables.

» Rising risk appetite & confidence support multiple expansion alongside earnings growth.

8.3 Explain why bond prices would fall on the back of stronger than expected retail sales. *(2)*

ANSWER

» Strong sales raise **inflation expectations** and the likelihood that the central bank will **hike interest rates**.

» Higher expected policy rates push **bond yields up**, and because bond **yields and prices move inversely**, bond **prices fall**.

QUESTION 9 · Monetary Policy & SARB [13 marks]

9.1 What is the repo rate and how does the lowering of this rate affect the local economy? *(4)*

ANSWER

The **repo rate** (repurchase rate) is the rate at which the **South African Reserve Bank lends short-term funds to commercial banks** against eligible collateral. It is the SARB's headline monetary policy lever.

Effect of a cut:

» Banks' wholesale funding becomes cheaper → they pass on lower lending rates (prime, mortgages, business credit).

» Cheaper credit stimulates **household consumption** and **business investment** → boosts aggregate demand and employment.

» Bond yields fall → bond prices rise; equity markets often rally on cheaper discount rates.

» Rand may **weaken** on narrower yield differentials, supporting exporters but lifting imported-inflation risk.

9.2 What other tools are available to a central bank wishing to implement expansionary monetary policy? Your answer should demonstrate your understanding of how each tool operates, its effects on market participants and how it works to encourage economic growth. *(4)*

ANSWER

» **Open Market Operations (OMO)** — SARB **buys** government bonds in the market, paying with newly created reserves. Bank liquidity rises, short-term rates fall, banks lend more freely → credit, spending and investment expand.

» **Reserve requirement cuts** — lowering the minimum % of deposits banks must hold at the central bank frees up cash for lending. Each rand of reserves supports more credit through the money multiplier.

» **Quantitative Easing (QE)** — large-scale purchases of long-dated government and sometimes corporate bonds beyond normal OMOs. Pushes **long-end yields** down, raises asset prices, encourages risk-taking and investment; deployed when the policy rate is at or near the zero lower bound.

» **Forward guidance** — explicit communication that rates will stay low (or be cut further). Anchors market expectations of future short rates, pulling longer-term yields down today and encouraging investment.

9.3 Describe the SARB's mandate in terms of inflation management. *(5)*

ANSWER

» **Constitutional mandate (s224):** protect the value of the currency in the interest of **balanced and sustainable economic growth**.

» Operates a **flexible inflation-targeting** framework with a target band of **3% to 6%** for headline CPI (with a 4.5% midpoint preferred).

» The **Monetary Policy Committee (MPC)** sets the repo rate to bring inflation back toward target over the policy horizon, while considering output and employment effects.

» **Operationally independent** from the executive — protected by legislation to ensure credibility and anchor inflation expectations.

» Reinforced by **transparency**: published MPC statements, Monetary Policy Reviews and inflation forecasts; uses the full monetary toolkit (repo, OMO, reserves, forward guidance) to support price stability.

Total marks: 110 | Q1: 15 · Q2: 9 · Q3: 9 · Q4: 24 · Q5: 12 · Q6: 15 · Q7: 8 · Q8: 5 · Q9: 13